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EX NO: 10	**5G Network Slicing Environment Using MATLAB**

Aim

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Software Required

• MATLAB

Theory:

Network slicing is a key feature of 5G that enables a single physical network infrastructure to be

partitioned into multiple logical networks called slices. Each slice is optimized for a specific

service category, such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency

Communications (URLLC), or massive Machine-Type Communications (mMTC). By

dynamically allocating network resources, network slicing improves flexibility, resource

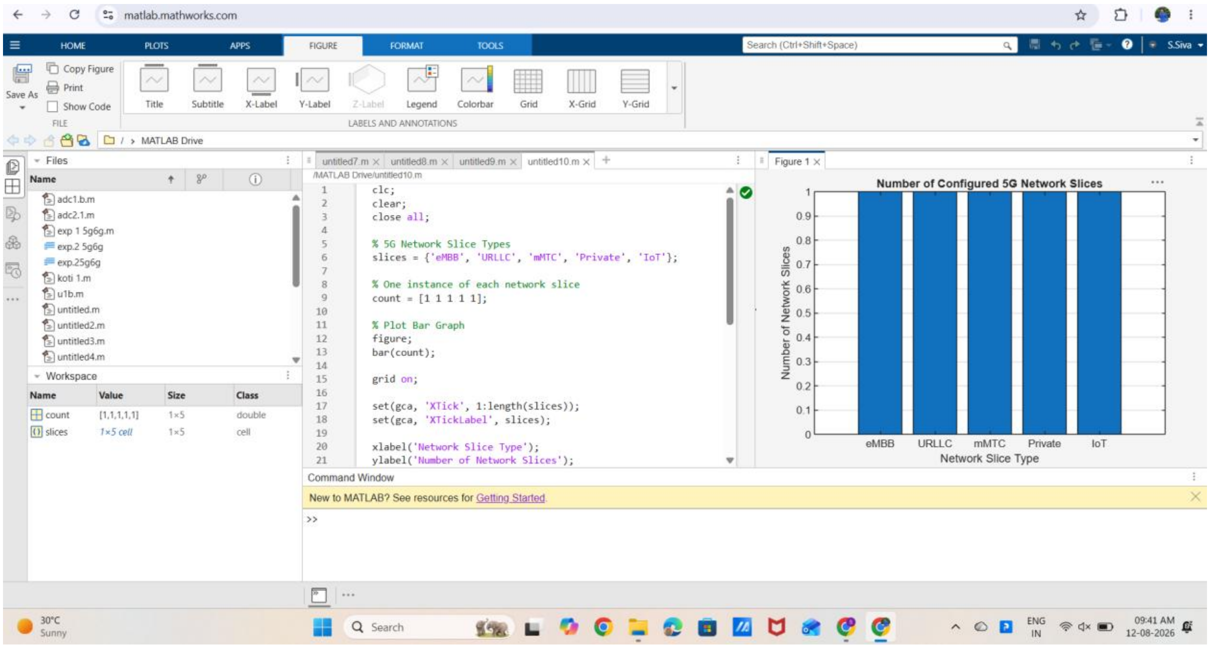
utilization, and Quality of Service (QoS) while supporting diverse application requirements.

MATLAB Code:

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};


count = [1 1 1 1 1]; % One instance of each network slice
```

```
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.